

CUDA Kernel Library

Driving a GEMM from naive to compute bound on an RTX 5070

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2026

Abstract

This report documents a CUDA kernel library whose general matrix multiply path is driven, through explicit optimization steps and Nsight Compute diagnostic rounds, from a naive baseline to a compute bound tensor core kernel. Every performance figure is my kernel measured against the vendor library (cuBLAS, or cuSPARSE for sparse matrix vector) on one GPU, an NVIDIA RTX 5070, in the same process. There are no cross device comparisons. The top FP16 kernel passes the compute bound gate and reaches 89.9 percent of cuBLAS at 4096 cubed and 88.9 percent at 8192 cubed, with a geometric mean of 79.9 percent over the four swept shapes. Alongside the GEMM ladder the library carries GEMV, TRSM, CSR SpMV, and a cuSOLVER wrapper, all verified against the vendor library, with a roofline whose compute ceilings come from the hardware rather than from a vendor library. The figures in this report were collected before the benchmark clocks were locked and before two timed region defects were fixed; the corrections register in `docs/CORRECTIONS.md` lists what that affects and what is pending re-measurement.

1 Introduction

The aim is narrow on purpose: take one operation, GEMM, and push a hand written kernel from a fair first pass to the point where Nsight Compute reports it as compute bound and it lands within about ten percent of cuBLAS on this specific card. The discipline that makes the result trustworthy is the comparison policy. Percent of cuBLAS on the same device is the only headline used, because it cancels the hardware out of the claim. No number in this report comes from anywhere but a run of this code on this machine, and each is traceable to a results file and a commit hash.

The machine is an RTX 5070 (Blackwell GB205, compute capability 12.0, sm_120, 48 SMs, 12 GB GDDR7), running CUDA Toolkit 13.3 inside WSL2. The ceilings used throughout are measured, not read from the datasheet: about 579 GB/s streaming bandwidth, 23 TFLOP/s FP32, and 65 TFLOP/s tensor core (best cuBLAS FP16).

2 Background

Roofline. The roofline model bounds attainable performance by $\min(\text{peak compute}, \text{peak bandwidth} \times \text{arithmetic intensity})$ [2]. The ridge, peak compute over peak bandwidth, separates the memory bound region (left) from the compute bound region (right). GEMM at large N has an arithmetic intensity that grows with N , so it lives far to the right of the ridge; a slow GEMM is therefore an implementation gap, not an algorithmic ceiling.

Blocking theory. The decisive step from a shared memory tiled kernel to a fast one is register blocking: each thread computes many outputs, which lifts the arithmetic intensity per thread and the work done per shared memory instruction. This follows Volkov and Demmel [1], and the diagnostic rounds below confirm it on Blackwell.

Tensor pipeline. The fifth generation tensor cores raise the compute ceiling far above the FP32 units. Reaching that ceiling means feeding the cores fast enough, which is a shared memory problem: the fragment loads, not the matrix multiply instructions, are what saturate first.

3 Methodology

Timing. Every measurement uses CUDA events with five warmup launches then twenty timed reps, reported as the median so a single slow rep does not move the headline. NVML samples power, clocks, temperature, and throttle reasons on a background thread during every sweep; a throttled run is rerun after cooldown rather than reported.

Diagnostic rounds. After each ladder rung, one Nsight Compute round: capture the Speed of Light, occupancy, memory workload, and warp stall reasons; name the top limiter; state a hypothesis; apply exactly one change; re-measure. Each round is one dated entry backed by an ncu report. This loop, not any single kernel, is the method.

4 GEMM design and optimization

The ladder exists so the benchmark table attributes each gain to a cause. Table 1 and Figure 1 give the measured result at 4096 cubed; the narrative below is the diagnostic log in brief.

rung	precision	GFLOP/s	percent of cuBLAS
naive	fp32	1908	8.7
shared tiled	fp32	1584	7.4
register blocked	fp32	10390	47.1
cp.async double buffered	fp32	16210	73.5
WMMA	fp16	36333	59.3
mma.sync PTX	fp16	36343	59.5
mma.sync + ldmatrix	fp16	36550	59.8
mma.sync + ldmatrix + swizzle	fp16	54985	89.9
WMMA (BF16)	bf16	33563	53.4

Table 1: GEMM ladder at 4096 cubed, each rung versus cuBLAS of the same precision, measured on the RTX 5070.

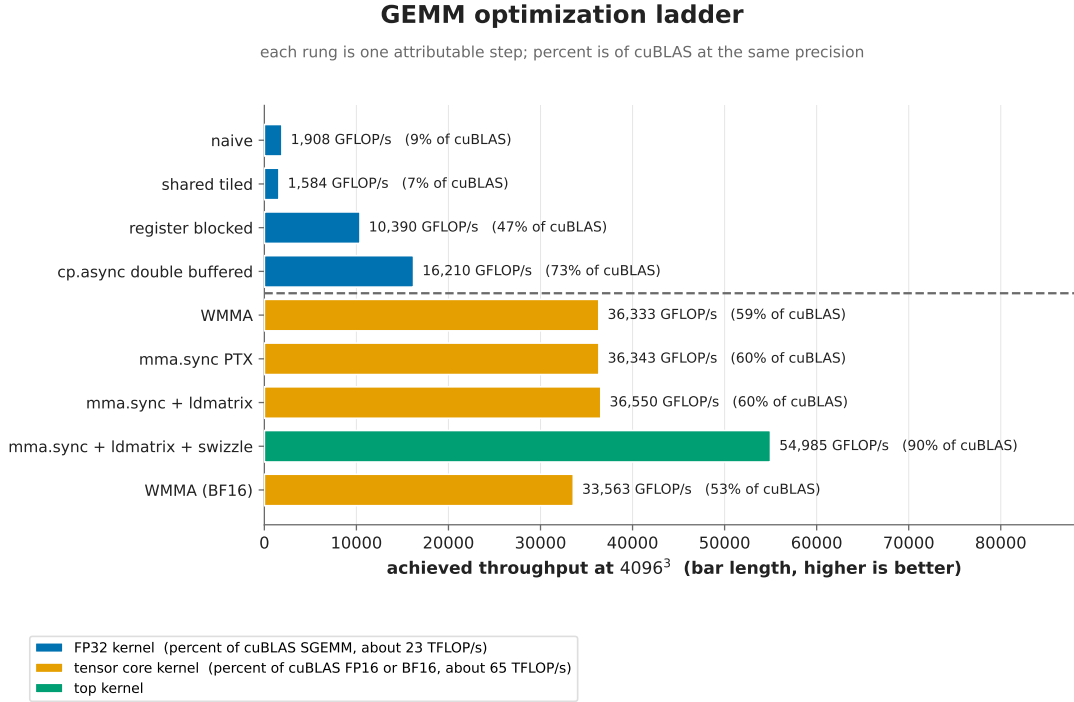


Figure 1: GEMM ladder at 4096 cubed. Bar length is the absolute achieved throughput (GFLOP/s). The percent in each label is of cuBLAS *at the same precision*, which is why a shorter FP32 bar can show a higher percent than a longer tensor bar: the FP32 kernels (blue) are measured against cuBLAS SGEMM (about 23 TFLOP/s) and the tensor kernels (amber, top kernel green) against cuBLAS FP16 or BF16 (about 65 TFLOP/s).

Naive to tiled (Round 1). The naive kernel reaches about 1.9 TFLOP/s. The shared memory tiled kernel, the textbook first optimization, is 17 percent slower than the kernel it was supposed to improve on (1584 against 1908 GFLOP/s at 4096 cubed), and that is the most interesting result in this paper: the move everyone reaches for first is a regression on this part. The round names the cause: the tiled 32 by 32 block is 1024 threads, so only one block fits per SM (about 66 percent occupancy against 100 percent for naive), and its top stall is the MIO (shared memory) pipe. Meanwhile the naive kernel is not bandwidth bound at all; its L1 hit rate is 87 percent and DRAM throughput only 37 percent, because the 48 MB L2 absorbs the redundant reads. Tiling alone does not win here.

Register blocking (Round 2). Giving each thread an 8 by 8 register tile lifts throughput to 10.4 TFLOP/s, 5.5 times naive and the decisive step. Warp cycles per issued instruction fall from 45 to 9. The new limiter is the shared read pipe and a 33 percent occupancy ceiling from 106 registers per thread.

cp.async double buffering (Round 3). Overlapping the next tile’s global load with the current tile’s math takes the FP32 kernel to 16.2 TFLOP/s, 73 percent of cuBLAS. The FP32 path is now occupancy limited (149 registers per thread, 17 percent occupancy) and is accepted as the top FP32 variant, its gap diagnosed.

Tensor cores (Rounds 4 and 5). Moving the inner product onto the tensor cores with WMMA doubles throughput to 36 TFLOP/s, but the cores are starved. In the Round 4 detail page ([experiments/results/ncu/round04/wmma_fp16_4096.txt](#)) the GPU Speed Of Light section reads Compute (SM) Throughput 53.72 percent against Memory Throughput 92.40 percent, and the row that actually names the culprit is L1/TEX Cache Throughput at 94.38 percent: the fragment loads saturate the shared read path while the tensor pipe idles. Dropping to mma.sync PTX with hand placed fragments changes nothing, which is the informative result: the instruction path was never the bottleneck, the fragment loads are.

Toward the gate (Rounds 6 to 9). ldmatrix on the small tile does not move the number either, because at a 64 by 64 tile the bytes to compute ratio is too high. A 128 by 128 tile with cp.async double buffering takes it to 80 percent of cuBLAS and flips the Speed of Light to co limited. A three stage pipeline is tried and reverted (it costs occupancy). The bank conflict counter then shows the real problem: 219 million shared load conflicts. A shared memory swizzle, applied identically to the stores and the loads so correctness is unconditional, cuts them to 33.7 million, collapses the memory Speed of Light from 76 to 30 percent, and lifts compute to 83 percent. The kernel now reaches 89.9 percent of cuBLAS at 4096 cubed and 88.9 percent at 8192 cubed, and passes the compute bound gate. It is not a flat 90 percent across the range: at 1024 cubed it is 63.5 percent and at 2048 cubed 80.5 percent, because the small shapes do not fill 48 SMs.

5 Supporting families

Each family is correct against the vendor library and benchmarked on the same device (Table 2). GEMV is a bandwidth benchmark in disguise: its arithmetic intensity is fixed near 0.5 FLOP per byte, so the warp and vectorized kernels reach cuBLAS and the measured streaming bandwidth while the naive kernel is coalescing limited. CSR SpMV is tested on a skewed degree matrix, where the warp per row kernel is about 1.6 times the thread per row kernel; its percent of cuSPARSE is

withdrawn, because the cuSPARSE wrapper built its descriptors and its workspace inside the timed region, so the denominator included setup my own kernels never paid. TRSM reduces most of its work to a GEMM style trailing update. The blocked kernel agrees with cuBLAS to about $1e-7$, which is a correctness result and not a performance one: on speed it reaches 381 GFLOP/s against cuBLAS STRSM’s 1200, 31.8 percent, and closing that gap is open work. The cuSOLVER LU and Cholesky solvers are checked by residual (about $5.6e-7$ and $4.3e-7$), not merely by the absence of an error.

kernel	shape	GFLOP/s	percent of vendor
GEMV warp	8192 sq	304.0	99.9
GEMV vectorized	8192 sq	308.1	101.6
CSR SpMV warp	131072	93.6	127.5
CSR SpMV naive	131072	57.5	86.1
TRSM blocked	2048	381.3	31.8
TRSM naive	2048	7.8	0.7

Table 2: Supporting families on the RTX 5070, each versus its vendor baseline.

6 Roofline analysis

Figure 2 places the kernels against the ceilings. The bandwidth ceiling is measured on this GPU; the compute roofs are the hardware’s, computed as 48 SMs times the SM clock times 512 FLOP per cycle for FP16 inputs with FP32 accumulate, and 256 FLOP per cycle for FP32. That distinction matters: setting the roof to a cuBLAS measurement, as an earlier version of this profiler did, turns every percent of roof into a percent of cuBLAS drawn on log axes. cuBLAS is drawn instead as a dashed attainable line, and it reaches 97.2 percent of the tensor roof, which is itself the best evidence that 512 FLOP per cycle is the right rate. The GEMM variants sit far to the right of both ridges, in the compute bound region; the top kernel lands at 84.1 percent of the tensor roof. GEMV sits at 0.5 FLOP per byte, on the bandwidth diagonal in the memory bound region. This is the evidence behind the compute bound claim and behind the tile choice. The roofline also settles one epilogue fusion decision: a standalone bias plus ReLU pass has an intensity of 0.167 FLOP per byte and is therefore memory bound, so folding it into the compute bound GEMM removes a pure bandwidth pass, measured at an 8.4 percent saving on the fused shape.

7 Discussion

The result that most shaped the work is that tiling alone loses to naive on this card, because the large cache hierarchy makes the naive kernel’s redundant traffic cheap. Register blocking, then feeding the tensor cores through a swizzled shared layout, are what actually move the number. The remaining ten percent to cuBLAS at the large shapes, and the much larger gap at 1024 cubed, is where its per shape tuning (deeper pipelines, split K for the small shapes that cannot fill 48 SMs) lives. The failed three stage pipeline round is kept in the engineering log as an equal rank result: it is the honest record that a plausible change measured worse and was reverted.

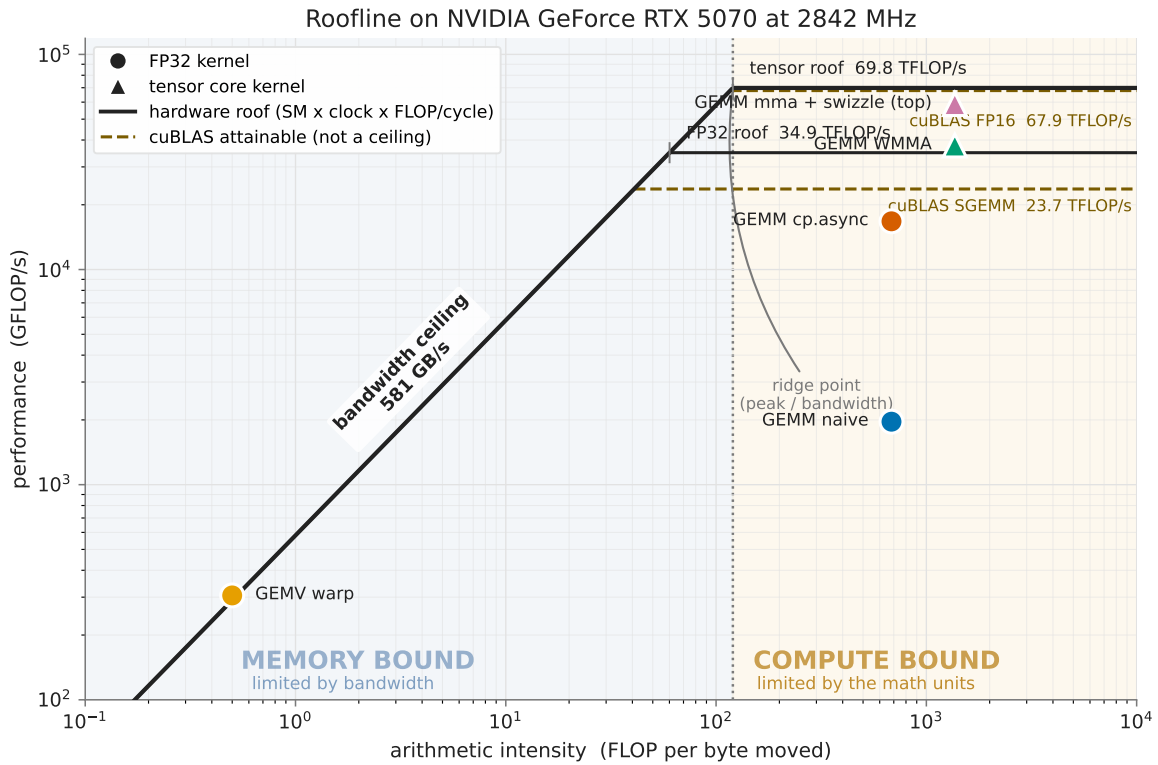


Figure 2: Roofline on the RTX 5070. The bandwidth ceiling is measured; the compute roofs are the hardware's, and cuBLAS is a dashed attainable line rather than a ceiling. GEMM is compute bound; GEMV is memory bound.

8 Conclusion

A hand written FP16 GEMM on the RTX 5070 reaches 89.9 percent of cuBLAS at 4096 cubed and is demonstrably compute bound, arrived at through nine logged diagnostic rounds with one change each. The supporting families are correct and benchmarked, the roofline is built on hardware ceilings, and every number is regenerated from the results files in the tree. The method, one change per round against profiler evidence, is the transferable part. What the corrections register in `docs/CORRECTIONS.md` adds to that method is the step I had skipped: auditing my own claims against my own data before publishing them.

References

- [1] Vasily Volkov and James W. Demmel. Benchmarking GPUs to tune dense linear algebra. In *Proceedings of the 2008 ACM/IEEE Conference on Supercomputing (SC)*, pages 1–11. IEEE Press, 2008.
- [2] Samuel Williams, Andrew Waterman, and David Patterson. Roofline: An insightful visual performance model for multicore architectures. *Communications of the ACM*, 52(4):65–76, 2009.